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Self-reported side effects of semaglutide and tirzepatide in online communities

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Social media can reveal patient experiences with glucagon-like peptide-1 receptor agonists (GLP-1 RAs) that extend beyond clinical trial data. We analysed 410,198 Reddit posts (May 2019–June 2025) mentioning semaglutide or tirzepatide. A total of 67,008 users self-reported using these medications, and 43.5% described at least one side effect. Gastrointestinal symptoms predominated, including nausea (36.9%), fatigue (16.7%), vomiting (16.3%), constipation (15.3%) and diarrhoea (12.6%). Notably, reproductive symptoms (for example, menstrual irregularities) and temperature-related complaints (for example, chills and hot flushes) emerged as unrecognized potential effects. These findings highlight patient concerns not well captured in current labelling or trials. Large-scale social media analysis can complement traditional pharmacovigilance by detecting emerging safety signals and expanding understanding of the real-world safety profile of GLP-1 RAs.

Two glucagon-like peptide-1 receptor agonists (GLP-1 RAs), semaglutide and tirzepatide, have rapidly gained popularity for type 2 diabetes management and weight control. Clinical trials and regulatory documents have established gastrointestinal symptoms, such as nausea, vomiting and diarrhoea, as common side effects of these medications. Social media data provide the opportunity to capture a broader range of patient experiences, including symptoms not formally recognized in drug labelling[1]. Understanding these real-world reports can help clinicians and regulators build a more comprehensive safety profile.

In this paper, we systematically characterize self-reported side effects of semaglutide and tirzepatide shared on Reddit, a social media platform with over 100 million daily active users anonymously posting in topical forums (subreddits)[2]. Reddit was selected as the data source for this study due to its large, publicly accessible, topic-specific health communities and its long-form, context-rich user discussions. Prior work has established Reddit as a valuable source for studying health-related disclosures, including sensitive or stigmatized medical experiences, and as an important resource for public health and computational social science research[3,4]. Unlike other major platforms, which provide either restricted, unreliable or prohibitively costly application programming interface access, publicly released researcher-curated corpora have made Reddit data practically feasible for large-scale public health research.

We conducted a cross-sectional study of self-reported side effects of semaglutide and tirzepatide using Reddit data from May 2019 through June 2025. Posts and comments were collected from nine large subreddits that discuss GLP-1 RAs or weight management. We first identified posts where users disclosed taking semaglutide or tirzepatide in any Food and Drug Administration-approved formulation for type 2 diabetes or weight management using a generative pre-trained transformer (GPT)-4o-mini-based classifier. This classifier also extracted the specific medications mentioned. Among posts indicating self-use, we applied a GPT-4.1-mini-based classifier to extract self-reported side effects and map them to Medical Dictionary for Regulatory Activities (MedDRA) Preferred Terms (PTs). Across the May 2019–June 2025 period during which eligible posts appeared, 172,679 of 410,198 Reddit posts mentioning semaglutide or tirzepatide indicated personal use by 67,008 users. Among them, 29,172 users (43.5%) reported at least one side effect, with a mean of 2.7 (standard deviation 2.4) PTs per user.

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Table 1 presents the prevalence of PTs among users reporting side effects. Nausea was most common (36.9%), followed by fatigue (16.7%), vomiting (16.3%), constipation (15.3%) and diarrhoea (12.6%). Other symptoms reported by ≥5% of users who disclosed side effects included decreased appetite, abdominal pain, gastroesophageal reflux, headache, abdominal distension and dizziness. The most frequent co-occurring PTs were nausea and vomiting, with 2,917 users (10.0%) self-reporting both (Supplementary Table 1).

Overall, 65.8% of semaglutide and/or tirzepatide users with side effects reported at least one gastrointestinal symptom. Psychiatric symptoms were noted by 12.9%, with anxiety (4.2%) and depression (2.8%) being most common.

As a secondary analysis, we examined monthly frequencies of the ten most commonly reported MedDRA PTs. Symptom counts increased in parallel with overall discussion volume but did not show distinct or interpretable temporal shifts (Supplementary Fig. 1).

As an exploratory analysis, we conducted a subanalysis of users who exclusively mentioned one formulation to determine differential side effect rates. Of the 29,172 users who self-disclosed at least one side effect, 17,937 (61.5%) exclusively mentioned semaglutide formulations and 7,125 (24.4%) exclusively mentioned tirzepatide formulations.

We then examined side effect frequencies separately for users who mentioned semaglutide exclusively (n = 17,937) versus tirzepatide (n = 7,125). Among semaglutide users reporting side effects, the most commonly reported symptoms were nausea (39.4%), fatigue (16.1%), vomiting (18.0%), constipation (14.9%) and diarrhoea (12.4%). Among tirzepatide users reporting side effects, the most commonly reported symptoms were nausea (28.6%), fatigue (14.7%), vomiting (11.1%), constipation (12.9%) and diarrhoea (12.5%). Injection site reactions, myalgia, insomnia and temperature-related symptoms (chills and feeling cold) were each reported by 1–4% of tirzepatide users. A complete descriptive summary for each formulation is provided in Supplementary Table 2.

In this large-scale Reddit analysis, over 67,000 semaglutide or tirzepatide users were identified, with 44% reporting side effects. Consistent with the findings of clinical trials, gastrointestinal symptoms were reported most commonly by these users[5–8]. The most frequently reported gastrointestinal effects were nausea, vomiting, constipation, diarrhoea and abdominal pain, which are the only adverse reactions featured as occurring more commonly than with placebo in the Food and Drug Administration-approved prescribing information for all five versions of these medications. Additional symptoms reported by ≥5% of users (for example, decreased appetite, eructation, distension and dizziness) have similarly been noted in trials as more common than placebo[5–8]. Headache, though common in trials, shows mixed placebo comparisons, while fatigue, Reddit's second most reported symptom, has met reporting thresholds in relatively few trials[5–8].

We also identified several side effects not previously reported with semaglutide or tirzepatide. Nearly 4% of Reddit users with side effects reported reproductive symptoms, notably menstrual changes like intermenstrual bleeding (0.9%), heavy bleeding (0.9%) and irregular cycles (0.7%). These rates would be higher in female-only samples. While weight loss may alleviate some reproductive issues, both weight gain and loss are linked to menstrual irregularities in women with obesity[9]. Further, GLP-1 RAs are thought to impact food intake, in part, by engaging receptors in the hypothalamus, a brain region that also plays a central role in menstrual cycle regulation[10]. Additionally, some users described symptoms possibly linked to altered temperature regulation (for example, chills, hot flushes and pyrexia), warranting further investigation in placebo-controlled trials or post-market data, particularly due to glucagon's associations with increased thermogenesis[10].

Strengths of our study include a large and contemporary sample, the focus on unprompted, self-disclosed experiences, the systematic classification of symptoms using MedDRA and the development of a straightforward large language model-based pipeline to detect emerging, potentially under-reported signals in a timely manner.

Several limitations should be noted. First, patients using weight loss subreddits may differ from the overall population prescribed GLP-1 RAs. Specifically, Reddit users tend to be younger, are more likely to be male and are disproportionately located in the USA[11,12]. The demographic composition of Reddit users and self-selection into health-related discussions limit the generalizability of our findings. Without cross-platform validation or comparison to structured pharmacovigilance databases, we cannot quantify the magnitude or direction of these biases. Although Reddit provides one of the largest and most active online health communities and remains among the few platforms with accessible, high-volume data, future work should extend this approach to other sources—such as specialized patient forums, clinical review platforms or any social media systems that restore stable application programming interface access—to assess the consistency of patient-reported symptom patterns across environments. Our results should be interpreted as hypothesis-generating signals that require confirmation through traditional surveillance systems and prospective studies in representative populations[13].

Second, because users were not prompted to disclose all side effects, we cannot estimate their true prevalence. Participation in health-related subreddits is voluntary, and individuals who experience more severe, persistent or distressing symptoms may be more likely to post about them than those with neutral or positive experiences. Even those who reported side effects may not have disclosed every symptom experienced, and individuals' beliefs about which symptoms are potentially attributable to the medication may have influenced their report. Further, earlier work has found that, while clinical trials often rely on self-reporting, gastrointestinal symptoms are often under-reported and inconsistently reported compared with validated measures for functional gastrointestinal disorders[14,15]. For these reasons, self-reported experiences cannot be used to determine true event frequency or causality. The symptoms discussed here may be attributable to weight loss itself, changes in comorbidities or other unrelated factors. Additionally, dosage, route of administration and treatment duration were inconsistently reported in posts and were not analysed. Our analysis prioritized breadth of symptom identification over temporal granularity. While we examined monthly trends in common symptoms (Supplementary Fig. 1), detailed assessment of symptom onset timing or dose–response relationships would require structured metadata (for example, prescription dates or dose titration schedules) rarely disclosed in social media posts. Examination of these variables would require data sources with more complete and structured medication histories, such as electronic health records or prospective registries. We were unable to determine when or how frequently these events occurred over the course of treatment or whether event occurrence depended on patient characteristics, such as the presence of type 2 diabetes. Results should be viewed as hypothesis-generating signals about patient-perceived symptoms rather than estimates of causal effects or true event rates.

Third, natural language processing can misclassify or overlook nuanced context, and adverse drug event extraction from social media texts remains an active area of research[16]. We validated the specific models deployed in this study against manually annotated samples and achieved high performance comparable to prior adverse event extraction systems[1,16]. While systematic benchmarking across large language model architectures represents an important direction for future methods research, our validation approach ensured reliable extraction for this application. Additionally, certain self-reported effects may not fit neatly into existing MedDRA concepts or may have been misclassified based on the user's description. For example, some events classified as 'gastroesophageal reflux' based on user terminology might have instead been recorded as 'dyspepsia' in the context of a clinical trial. In our exploratory subanalysis restricted to users whose

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Table 1 | Frequency of MedDRA PTs among 29,172 Reddit users taking GLP-1 RAs

MedDRA term	No. of users self-reporting	Percentage of users with self-disclosed PT
Gastrointestinal disorders	19,185	65.8
Nausea	10,764	36.9
Vomiting	4,749	16.3
Constipation	4,463	15.3
Diarrhoea	3,686	12.6
Abdominal pain	2,493	8.5
Eructation	2,010	6.9
Gastroesophageal reflux disease	1,877	6.4
Abdominal distension	1,488	5.1
Flatulence	1,055	3.6
Abdominal discomfort	900	3.1
Gastrointestinal disorder	883	3.0
Dyspepsia	662	2.3
Retching	350	1.2
Dry mouth	321	1.1
Breath odour	187	0.6
Gastrointestinal motility disorder	167	0.6
Impaired gastric emptying	167	0.6
Dysphagia	142	0.5
General disorders and administration site conditions	8,395	28.8
Fatigue	4,870	16.7
Asthenia	738	2.5
Early satiety	666	2.3
Pain	619	2.1
Malaise	589	2.0
Chills	332	1.1
Injection site pain	283	1.0
Hunger	220	0.8
Feeling cold	183	0.6
Illness	156	0.5
Thirst	147	0.5
Injection site reaction	144	0.5
Pyrexia	139	0.5
Nervous system disorders	5,704	19.6
Headache	1,784	6.1
Dizziness	1,445	5.0
Somnolence	492	1.7
Dysgeusia	398	1.4
Syncope	346	1.2
Tremor	333	1.1
Lethargy	311	1.1
Paraesthesia	301	1.0
Migraine	294	1.0
Cognitive disorder	267	0.9
Burning sensation	190	0.7
Taste disorder	177	0.6
Metabolism and nutrition disorders	5,205	17.8
Decreased appetite	3,371	11.6
Hypoglycaemia	600	2.1
Increased appetite	549	1.9
Dehydration	473	1.6
Food aversion	365	1.3
Food craving	171	0.6

Table 1 (continued) | Frequency of MedDRA PTs among 29,172 Reddit users taking GLP-1 RAs

MedDRA term	No. of users self-reporting	Percentage of users with self-disclosed PT
Psychiatric disorders	3,756	12.9
Anxiety	1,230	4.2
Insomnia	901	3.1
Depression	827	2.8
Panic attack	280	1.0
Irritability	257	0.9
Anhedonia	191	0.7
Libido decreased	161	0.6
Mood swings	159	0.5
Apathy	149	0.5
Sleep disorder	148	0.5
Depressed mood	142	0.5
Skin and subcutaneous tissue disorders	2,552	8.7
Alopecia	914	3.1
Hyperhidrosis	362	1.2
Pruritus	335	1.1
Rash	193	0.7
Acne	183	0.6
Dermatitis allergic	166	0.6
Urticaria	149	0.5
Musculoskeletal and connective tissue disorders	1,833	6.3
Myalgia	540	1.9
Arthralgia	336	1.2
Back pain	325	1.1
Muscle atrophy	194	0.7
Muscle spasms	158	0.5
Musculoskeletal pain	134	0.5
Reproductive system and breast disorders	1,142	3.8
Intermenstrual bleeding	266	0.9
Heavy menstrual bleeding	261	0.9
Menstruation irregular	211	0.7
Menstrual disorder	178	0.6
Dysmenorrhoea	168	0.6
Investigations	832	2.9
Weight increased	415	1.4
Cardiac disorders	718	2.5
Tachycardia	512	1.8
Palpitations	205	0.7
Respiratory, thoracic and mediastinal disorders	673	2.3
Dyspnoea	169	0.6
Vascular disorders	628	2.2
Hypotension	166	0.6
Hot flush	151	0.5
Infections and infestations	453	1.6
Eye disorders	388	1.3
Visual impairment	218	0.7
Renal and urinary disorders	357	1.2
Injury, poisoning and procedural complications	234	0.8
Contusion	157	0.5

PTs are grouped by primary system organ classes (bold) in descending order. PTs reported by less than 0.5% of users and system organ classes reported by less than 1.0% are omitted for brevity. See Supplementary Table 3 for PTs ordered purely by frequency and Supplementary Tables 4 and 5 for frequencies of corresponding MedDRA high-level terms and high-level group terms.

posts exclusively mentioned one formulation, we present descriptive comparisons of side effect frequencies but do not make formal statistical inferences about medication-specific effects. Without demographic and clinical data to match or adjust for confounding between medication groups, we cannot determine whether observed differences in complaint frequencies are attributable to the formulations themselves or to differences in the populations using each formulation. Such analyses would require data sources with detailed patient characteristics to enable appropriate comparison groups. Future work should stratify symptom patterns by medication using data sources with richer clinical context and validated methods for establishing medication-specific causal effects.

Despite these limitations, social media data provide important insights into symptoms that are of concern to semaglutide and tirzepatide users and may be helpful in identifying rarer or under-reported effects that warrant further investigation. Online health communities are increasingly influential in shaping patient expectations and treatment decisions, and a growing body of research examines these platforms across a range of contexts[17–21]. Future work could extend this framework by examining more detailed temporal dynamics in symptom patterns and how they vary by medication type or formulation; incorporating severity distinctions in patient-reported symptoms; and evaluating multilingual or non-Reddit digital communities to understand cross-platform and cross-cultural variation in patient experiences.

Capturing these patient-reported experiences can alert clinicians and researchers to emerging patient concerns related to GLP-1 RA use. Our data suggest that clinicians may encounter patient inquiries about menstrual irregularities and body temperature fluctuations in addition to commonly reported gastrointestinal symptoms. We recommend that future prospective trials and post-market surveillance studies systematically assess fatigue and reproductive symptoms, particularly menstrual changes in female participants, using validated instruments rather than relying solely on passive adverse event reporting. These findings may also inform updates to patient counselling materials and regulatory labelling to address symptoms of concern better to patients in real-world settings. We encourage researchers to examine whether these patient-reported concerns occur more frequently with semaglutide and/or tirzepatide than with placebo.

Methods

Posts and comments were collected from nine large subreddits (forums) that discuss GLP-1 RAs or weight management. These subreddits were selected to align with those analysed by prior researchers who estimated real-world weight loss outcomes of these medications[22]. Relevant posts and comments containing mentions of semaglutide or tirzepatide were identified using a list of brand and generic names and common misspellings. Data were collected for the period 1 January 2015 to 30 June 2025; in practice, all posts that met inclusion criteria were dated May 2019–June 2025. All data were collected using Pushshift and Arctic Shift (an extension of Pushshift)[23,24]. Please see the Supplementary Methods for subreddits used and a full list of search terms.

We first identified posts in which users personally disclosed taking either semaglutide or tirzepatide in any formulation currently approved by the Food and Drug Administration for the treatment of type 2 diabetes or weight management using a GPT-4o-mini based text classifier (prompt in Supplementary Methods). The classifier also extracted the specific medications being taken. The classifier's self-use performance was evaluated against a manually annotated set of 100 randomly selected posts and had a sensitivity of 86% and specificity of 96%. Only posts that met the self-disclosure criterion were included in subsequent analyses.

We evaluated the model's medication extraction accuracy on a random sample of 100 posts containing 109 medication mentions. The model achieved a precision of 95% and recall of 98% (F1 = 96%) at the medication level. Common errors included extracting a specific medication when the post did not clearly state which drug was being used, often inferring the medication from the subreddit name (for example, assuming Ozempic for posts in r/Ozempic) rather than from explicit mentions in the post content ($n = 4$), and missing secondary (non-semaglutide or non-tirzepatide) medications ($n = 2$). At the post level, 93.0% of posts had all medications correctly extracted.

Among posts indicating self-use, we applied a second classifier, based on the GPT-4.1-mini model, to extract self-reported side effects. Prior work has shown that both classical and transformer-based natural language processing systems can map free text to standardized vocabularies such as MedDRA[25–27]. Our pipeline follows this normalization paradigm but uses a large language model classifier to directly map side effects mentioned in the user-generated text to standardized MedDRA PTs using retrieval augmented generation (prompt in Supplementary Methods). Manual evaluation of 100 randomly selected posts indicated that the classifier had a sensitivity (recall) of 97% and a positive predictive value (precision) of 87% for identifying the presence or absence of individual side effects. Because true negatives cannot be enumerated meaningfully for named entity recognition models at the span level, specificity is undefined for this granularity. The classifier extracted 2,013 unique side effect phrases, which were matched to existing MedDRA PTs via exact matching and manual review when needed.

Percentages were calculated based on the population of users who disclosed at least one side effect. Because users were not prompted to disclose side effects, we cannot estimate their true prevalence. Further, uncontrolled self-reported experiences cannot be used to determine causality. The symptoms discussed here may be attributable to weight loss itself, changes in comorbidities or other unrelated factors.

For users mentioning one formulation exclusively (either semaglutide or tirzepatide but not both), we conducted exploratory descriptive comparisons of side effect frequencies. For each MedDRA PT, we calculated proportions in each medication group to characterize the distribution of reported symptoms. These comparisons are presented descriptively without formal statistical testing, as we lack the demographic and clinical data necessary to adjust for potential confounding factors that may differ between users of each medication.

GPT-4o-mini was chosen for self-disclosure classification because, at the time that component of the study was conducted, it was among the most capable lightweight models available for large-scale processing at feasible cost. Midway through the project, OpenAI released GPT-4.1-mini with documented performance improvements over GPT-4o-mini across a range of tasks including instruction following (https://openai.com/index/gpt-4-1/), and we therefore used GPT-4.1-mini for identifying mentions of side effects and mapping them to MedDRA terms. Larger, reasoning and non-mini models (for example, GPT-4o or GPT-4.1) were not employed for cost reasons.

All analyses were conducted in Python 3.11 using standard data processing libraries. Data were collected from publicly accessible Reddit posts and comments using Pushshift and Arctic Shift during the study period. No attempts were made to access private or password-protected content. As data were public and non-identifiable, institutional review board review was not required. Human Ethics and Consent to Participate declarations were not applicable.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this study are publicly available from Reddit.com or are accessible via Pushshift and Arctic Shift. Due to Reddit's Terms of Use, we are unable to share the raw data. All Reddit data in this study were used in accordance of Reddit's Terms of Use.

Code availability

Code to reproduce the analyses can be found at https://github.com/sehgal-neil/glp1-side-effects-analysis.

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Author contributions

N.K.R.S.: conceptualization, methodology, formal analysis, investigation, data curation and writing—original draft. J.S.T.: conceptualization, methodology and writing—review and editing. L.U.: conceptualization, methodology and writing—review and editing. S.C.G.: conceptualization, methodology, writing—review and editing, and supervision.

Competing interests

J.S.T. reports receiving an investigator-initiated grant, on behalf of the University of Pennsylvania, from Novo Nordisk and receiving consulting fees from Currax Pharmaceuticals, LLC. The other authors declare no competing interests.

Additional information

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☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	All data used in this study are publicly available from Reddit.com, or are accessible via Pushshift and Arctic Shift.
Data analysis	All data was analyzed using Python version 3.1

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data used in this study are publicly available from Reddit.com, or are accessible via Pushshift and Arctic Shift.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	NA
Reporting on race, ethnicity, or other socially relevant groupings	NA
Population characteristics	NA
Recruitment	NA
Ethics oversight	As data were public and non-identifiable, IRB review was not required.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Quantitative, observational cross-sectional study using publicly available social media text. We analyzed Reddit posts and comments to identify self-disclosed use of semaglutide or tirzepatide and to extract self-reported side effects mapped to MedDRA Preferred Terms. Analyses are descriptive (prevalence of reported PTs among users who disclosed ≥1 side effect; plus descriptive comparisons by medication formulation among users who mentioned only one formulation).
Research sample	Public Reddit posts/comments from nine subreddits focused on GLP-1 medications and weight/weight loss communities: r/mounjaro, r/ozempic, r/fasting, r/intermittentfasting, r/keto, r/loseit, r/Semaglutide, r/SuperMorbidlyObese, r/PlusSize. The analytic population consists of unique Reddit users who self-disclosed taking semaglutide or tirzepatide. No individual-level demographics (age/sex/gender/race/location) were reliably available and were not used for inclusion/stratification. The sample is not representative of medication users in the population; it reflects self-selected Reddit participants.
Sampling strategy	Convenience sample of posts/comments drawn from the specified subreddits within the study window. Posts/comments were retrieved using keyword matching on brand/generic names and common misspellings, then filtered using an LLM-based classifier to identify self-disclosed medication use; among self-use posts, a second LLM-based classifier extracted side effects and mapped them to MedDRA PTs. All eligible posts within the collection window were included.
Data collection	Reddit posts/comments accessed via Pushshift and Arctic Shift for the collection period. Data consist only of public text content and associated timestamps/metadata available in the dataset.
Timing	Posts/comments collected for January 1, 2015 through June 30, 2025; in practice, all posts meeting inclusion criteria were dated May 2019 through June 2025.
Data exclusions	Exclusions were applied at the post/comment level based on content and eligibility: 1. Posts/comments that mentioned the medications but did not contain a self-disclosure of personal use were excluded from downstream side-effect analyses. 2. Users were included in side-effect prevalence calculations only if they self-disclosed at least one side effect. 3. For the medication-comparison sub-analysis, users who mentioned both semaglutide and tirzepatide were excluded (restricted to users exclusively mentioning one formulation). Across the May 2019–June 2025 period during which eligible posts appeared, 172,679 of 410,198 Reddit posts mentioning semaglutide or tirzepatide indicated personal use by 67,008 users. Among them, 43.5% (29,172 users) reported at least one side effect. In exploratory descriptive analyses, we examined side effect frequencies separately for users who exclusively mentioned semaglutide (n=17,937) versus tirzepatide (n=7,125).

Non-participation	NA
Randomization	NA

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging